

Jiashuai (Joshua) Chai

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Portfolio: <https://joshuachai.github.io/PersonalCV/>

Summary

Junior software developer with internship and team project experience in Australia. Worked on web applications using React and Django and developed Python image analysis software for medical research. Seeking a developer role to contribute to projects and learn from experienced developers.

Technical Knowledge

- Languages: JavaScript, TypeScript, HTML, CSS, Python, R
- Frameworks & Libraries: React, Next.js, Tailwind CSS, Django
- Development Tools: Docker, VS Code, Codex, Claude Code, Git, GitHub

Education

Master of Professional Engineering (Software Engineering)

Jul 2024 – Jul 2026

The University of Western Australia, Perth, Australia

Bachelor of Computer Science and Technology

Oct 2020 – Jun 2024

Heilongjiang University of Science and Technology, Harbin, China

Employment History

Lions Eye Institute, Perth, Australia

Feb 2025 - Jun 2026

Software Engineer Intern

Feb 2025 - May 2025; Nov 2025 - Feb 2026

- Automating image formatting and file renaming using Python to prepare datasets for processing.
- Migrating components of an existing image-analysis application from MATLAB to Python.
- Implementing calculation logic and integrating results into the application's user interface.
- Debugging processing workflows and updating features based on researcher feedback.

Achievement: Reduced processing time from around 60 seconds to under 30 seconds by optimising processing algorithms.

Research Assistant, UWA Thesis Project

Jun 2025 - Oct 2025; Feb 2026 - Jun 2026

- Continuing to develop OCTAVA as part of a master's thesis project, building on previous internship work.
- Reviewing research needs and existing approaches to define the requirements for a new software module.
- Developing a new analysis module within the existing Python application while maintaining its original functionality.
- Updating the user interface with adjustable settings and controls for selecting analysis regions and viewing results.
- Reusing existing application components and integrating the new module with result displays and file exports.
- Comparing alternative algorithms using test data and testing the complete workflow from input to export.

Achievement: Delivered the new OCTAVA analysis module and completed the master's thesis and final defence.

Personal Attributes

- **Communication:** Discussed requirements with clients and colleagues, explaining design decisions and demonstrating software features during weekly team meetings.
- **Collaboration:** Worked with other developers to complete application features by connecting frontend interfaces to backend APIs and resolving merge conflicts together.
- **Adaptability:** Learned unfamiliar medical concepts through independent research to understand project requirements and how the software would support researchers' work.

Volunteer Work

International Student Ambassador

Dec 2024 – Aug 2026

StudyPerth, Perth, Australia

Delivered orientation presentations to new international students and worked with staff on student engagement activities.

Student Advisory Committee Member

Jul 2025 - Oct 2025

The McCusker Centre for Citizenship, Perth, Australia

Shared internship experiences during alumni panel presentations and provided individual guidance to students at events.

References

Available upon request.